

PORTAGE GLACIER, AK, USA

WMO: 700001

Lat: 60.785N

Lon: 148.841W

Elev: 103

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 26492

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-7.3	-18.2	2.0	-12.9	-11.7	2.9	-6.0	32.5	39.9	28.5	38.6	1.0	300	0.749

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	69.3	58.2	65.8	55.8	63.2	54.5	59.0	67.4	57.0	64.1	55.3	61.2	7.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.4	63.3	61.6	53.4	60.8	59.5	52.1	58.1	58.0	25.6	67.5	24.4	64.4	23.4	61.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	23.7	19.8	-17.1	73.4	6.5	4.6	-21.8	76.7	-25.5	79.4	-29.2	82.0	-33.8	85.3		
(5)			-17.4	60.9	6.4	2.2	-22.0	62.5	-25.8	63.8	-29.4	65.0	-34.0	66.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	22.7	24.8	26.4	35.6	44.2	51.1	54.6	52.8	47.2	37.7	25.9	25.1		
	DBStd	14.70	14.88	11.58	10.56	5.70	4.50	3.75	3.69	3.34	4.58	7.43	12.47	12.55		
	HDD50	4924	846	705	730	432	187	30	5	10	102	382	723	772		
	HDD65	10066	1311	1125	1195	882	643	417	322	379	535	847	1173	1237		
	CDD50	333	0	0	0	0	8	63	148	96	17	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	21	0	0	0	0	0	0	15	6	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	7.9	8.3	8.9	7.6	8.2	8.1	7.4	7.4	7.1	7.7	7.7	7.7	8.5		
	PrecAvg	149.9	13.4	12.7	11.3	11.9	10.5	6.9	8.6	11.3	17.1	16.2	15.4	15.8		
(15) Precipitation	PrecMax	179.2	35.9	23.5	22.1	25.7	24.6	11.5	15.5	22.7	36.4	34.0	39.5	29.8		
	PrecMin	87.0	2.1	1.6	2.3	1.8	1.5	0.8	4.6	3.5	3.9	4.8	1.1	4.5		
	PrecStd	20.6	7.4	6.1	5.5	5.7	6.8	2.6	3.0	5.2	7.8	6.5	10.1	6.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	45.2	47.8	54.5	65.9	71.1	76.2	73.3	63.0	53.9	46.7	45.0		
		MCWB	41.9	38.0	39.8	45.0	51.7	58.7	62.4	59.8	53.2	48.3	42.3	40.8		
	2%	DB	42.5	40.9	43.2	49.2	61.8	66.4	70.1	66.9	59.2	50.8	44.7	42.8		
		MCWB	39.6	37.9	39.0	41.1	49.4	55.1	59.1	56.7	51.4	46.6	41.5	39.2		
	5%	DB	40.7	39.2	40.9	46.1	58.3	63.9	66.3	63.8	56.3	48.6	42.8	40.8		
		MCWB	38.0	36.6	37.5	39.3	47.5	54.2	57.1	55.2	50.5	44.7	40.0	37.8		
	10%	DB	39.1	37.6	38.6	43.6	54.5	60.9	63.4	60.8	54.3	47.0	40.6	39.1		
		MCWB	36.8	35.1	35.6	38.2	45.9	52.2	55.7	53.7	49.7	43.4	38.2	36.7		
	0.4%	WB	42.4	39.6	41.5	45.6	52.8	60.2	62.5	61.2	55.0	49.9	44.1	41.4		
		MCDB	45.6	42.6	45.3	53.5	64.3	68.4	74.2	71.4	59.2	52.6	45.6	43.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.2	38.2	39.5	42.1	50.0	56.8	59.6	57.9	53.0	47.7	42.0	39.7		
		MCDB	42.3	40.8	42.7	47.4	59.7	65.1	68.7	65.5	57.1	50.3	44.2	42.4		
	5%	WB	38.3	36.9	37.5	40.3	48.2	54.5	57.7	55.9	51.5	45.9	40.2	38.3		
		MCDB	40.3	39.1	40.0	44.6	56.6	62.2	65.1	61.8	54.8	48.2	42.5	40.6		
	10%	WB	36.9	35.5	36.0	39.0	46.7	52.7	56.1	54.4	50.5	44.1	38.3	37.0		
		MCDB	38.8	37.2	38.1	42.7	53.3	59.8	62.3	59.1	53.6	46.5	40.3	38.9		
	5% DB	MDBR	10.4	11.0	14.4	13.0	14.3	14.6	13.8	13.7	12.8	10.1	9.8	10.3		
		MCDBR	7.6	7.5	11.5	19.0	26.4	24.8	23.9	23.9	18.5	10.9	8.0	7.8		
		MCWBR	5.5	5.3	7.8	11.9	15.1	14.8	14.1	14.8	12.3	7.6	5.8	5.6		
		MCWBR	7.4	7.3	9.5	15.1	21.9	22.4	21.6	19.2	15.5	9.2	7.6	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.310	0.324	0.369	0.381	0.384	0.388	0.372	0.322	0.310	0.253	0.238		
		taud	2.053	2.064	2.076	2.015	2.116	2.246	2.307	2.352	2.478	2.274	2.042	1.912		
	Ebn at Noon		152	204	247	257	265	267	263	258	255	208	143	92		
		Edh at Noon	13	25	34	44	43	39	36	32	23	20	13	10		
(44) All-Sky Solar Radiation	RadAvg		96	315	815	1332	1660	1656	1492	1177	775	360	138	53		
	RadStd		13	49	95	136	142	90	118	131	111	43	22	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	+3.42	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page